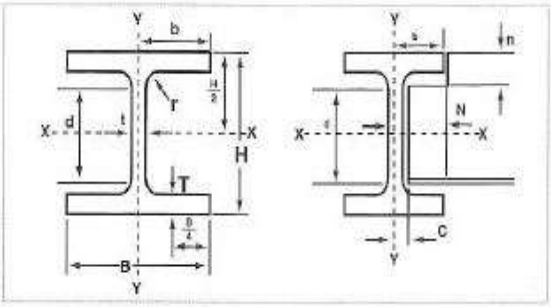
Dimensions and Properties

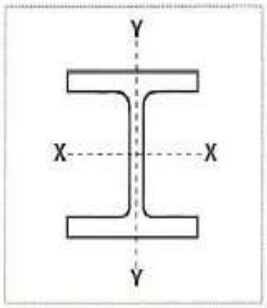


4a (ii) Columns (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Column	Width Of Column	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W4	4 x 4	(102 x 102)	13	19.3	105.7	103.1	8.8	7.1	6.4	75.5	24.7	5.88	10.6	6	58	15
			13.8	20.5	102.0	102.0	9.4	6.8	6.0	71.2	26.1	5.43	8.90	6	57	15
			16.3	24.3	107.0	100.0	12.0	7.9	6.0	71.0	30.9	4.17	8.99	6	56	18
W5	5 x 5	(127 x 127)	16	23.8	127.3	127.0	9.1	6.1	7.6	93.8	30.4	6.95	15.4	5	70	17
			19	28.3	130.8	127.8	10.9	6.9	7.6	93.8	35.8	5.86	13.7	5	70	19
W6	6 x 6	(152 x 152)	15	22.3	152.1	152.1	6.6	5.8	6.4	126.2	28.5	11.5	21.6	5	83	13
			16	23.8	152.4	152.2	6.8	5.8	7.6	123.6	29.2	11.2	21.3	5	83	14
			20	29.8	157.6	152.9	9.4	6.5	7.6	123.6	38.3	8.13	19.0	5	83	17
			25	37.2	161.8	154.4	11.5	8.0	7.6	123.6	47.1	6.71	15.5	6	83	19
W8	8 x 8	(203 x 203)	31	46.1	203.2	203.6	11.0	7.2	10.2	160.8	58.7	9.25	22.3	6	108	21
			35	52.1	206.2	204.3	12.5	7.9	10.2	160.8	66.3	8.17	20.4	6	108	23
			40	59.5	209.6	205.8	14.2	9.4	10.2	160.8	76.4	7.25	17.1	7	108	24
			48	71.4	215.8	206.4	17.3	10.0	10.2	160.8	90.4	5.97	16.1	7	108	28
			58	86.3	222.2	209.1	20.5	12.7	10.2	160.8	110	5.10	12.7	8	108	31
			67	99.7	228.6	210.3	23.8	14.5	10.2	160.7	127	4.43	11.1	9	108	34
W10	10 x 10	(254 x 254)	49	72.9	254.1	254.6	14.2	8.6	12.7	200.3	93.1	8.96	23.3	6	133	27
			54	80.4	256.3	254.8	15.6	9.4	12.7	199.7	102	8.17	21.2	7	133	28
			60	89.3	260.3	256.3	17.3	10.3	12.7	200.3	113	7.41	19.4	7	133	30
			68	101.2	264.2	257.3	19.6	11.9	12.7	199.7	129	6.58	16.7	8	133	32
			72	107.1	266.7	258.8	20.5	12.8	12.7	200.3	136	6.31	15.6	8	133	33
			77	114.6	269.2	258.8	22.1	13.5	12.7	199.6	146	5.86	14.8	9	133	35
			88	131.0	275.3	260.7	25.2	15.4	12.7	199.6	167	5.18	13.0	10	133	38
			89	132.4	276.3	261.3	25.3	15.3	12.7	200.3	168	5.16	13.1	10	133	38
			100	148.8	281.9	262.6	28.5	17.3	12.7	199.6	190	4.62	11.6	11	133	41
			112	166.7	289.1	265.2	31.7	19.2	12.7	200.3	213	4.18	10.4	12	133	44
W12	12 x 12	(305 x 305)	65	96.7	307.9	305.3	15.4	9.9	15.2	246.7	123	9.91	24.9	7	158	31
			72	107.1	311.1	305.8	17.0	10.9	15.2	246.7	136	8.98	22.6	7	157	32
			79	117.6	314.5	307.4	18.7	12.0	15.2	246.7	150	8.22	20.6	8	158	34
			87	129.5	318.3	308.0	20.6	13.1	15.2	246.8	165	7.49	18.9	9	157	36
			92	136.9	320.5	309.2	21.7	13.8	15.2	246.7	174	7.12	17.9	9	158	37
			96	142.9	322.8	308.9	22.9	14.0	15.2	246.7	182	6.76	17.7	9	157	38
			106	157.7	327.1	311.2	25.0	15.8	15.2	246.7	201	6.22	15.6	10	158	40
			120	178.6	333.2	312.9	28.1	18.0	15.2	246.7	228	5.57	13.7	11	157	43
			133	197.9	339.9	314.5	31.4	19.1	15.2	246.7	252	5.01	12.9	12	158	47
			136	202.4	340.6	315.0	31.8	20.1	15.2	246.7	258	4.96	12.3	12	157	47
			152	226.2	348.2	317.0	35.6	22.1	15.2	246.7	289	4.46	11.2	13	157	51
			161	239.6	352.5	318.4	37.7	23.0	15.2	246.7	306	4.22	10.7	14	158	53
			170	253.0	356.4	319.3	39.6	24.4	15.2	246.8	323	4.03	10.1	14	157	55
			190	282.8	365.3	322.2	44.1	26.8	15.2	246.7	360	3.65	9.21	15	158	59
			210	312.5	373.6	324.9	48.3	30.0	15.2	246.7	399	3.37	8.23	17	157	63
			230	342.3	382.3	327.5	52.6	32.6	15.2	246.7	437	3.11	7.57	18	157	68
			252	375.0	391.4	330.3	57.2	35.4	15.2	246.6	478	2.89	6.97	20	157	72
			278	413.7	402.6	333.8	62.7	38.9	15.2	246.7	529	2.66	6.35	21	157	78
			279	415.2	402.6	333.8	62.7	38.9	15.2	246.8	528	2.66	6.34	21	157	78
			305	453.9	414.5	336.2	68.7	41.3	15.2	246.7	578	2.45	5.97	23	157	84
			336	500.0	427.2	340.0	75.1	45.1	15.2	246.6	638	2.26	5.47	25	157	90

Note: Calculated based on 1kg = 2.2046 lb and 1 m = 3.2808 feet

Dimensions and Properties



4a (ii) Columns (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J		
lb/ft	kg/m	m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3			dm^6	cm^4	in	
13	19.35	0.599	472	160	4.38	2.55	89.4	31.1	103	47.8	0.836	10.9	0.0038	6.29	**4 x 4**	**W4**
13.8	20.54	0.586	456	167	4.18	2.53	89.4	32.7	104	50.4	0.827	9.61	0.0036	7.78		
16.3	24.26	0.588	587	200	4.36	2.55	110	40.1	129	61.5	0.844	8.13	0.0045	13.5		
16	23.81	0.737	892	312	5.42	3.21	140	49.2	158	75.0	0.842	13.0	0.0109	8.00	**5 x 5**	**W5**
19	28.28	0.746	1092	380	5.52	3.25	167	59.4	190	90.6	0.845	11.2	0.0136	13.1		
15	22.32	0.890	1210	387	6.51	3.68	159	50.9	177	77.7	0.837	21.4	0.0205	4.21	**6 x 6**	**W6**
15.7	23.36	0.889	1250	400	6.54	3.70	164	52.6	182	80.2	0.840	20.7	0.0212	4.63		
20	29.76	0.901	1748	560	6.76	3.83	222	73.3	248	112	0.849	16.0	0.0308	10.5		
25	37.20	0.912	2210	706	6.85	3.87	273	91.5	309	140	0.848	13.3	0.0399	19.2		
31	46.13	1.19	4568	1548	8.82	5.13	450	152	497	231	0.847	17.7	0.143	22.2	**8 x 8**	**W8**
35	52.09	1.20	5259	1778	8.91	5.18	510	174	567	264	0.848	15.8	0.167	31.8		
40	59.53	1.21	6125	2065	8.96	5.20	584	201	656	305	0.846	14.1	0.197	47.2		
48	71.43	1.22	7618	2537	9.18	5.30	706	246	799	374	0.853	11.9	0.250	80.2		
58	86.31	1.24	9449	3128	9.28	5.34	850	299	977	456	0.850	10.2	0.318	137		
67	99.71	1.25	11310	3687	9.44	5.39	990	351	1150	536	0.852	9.0	0.387	211		
49	72.92	1.49	11410	3908	11.1	6.48	898	307	992	465	0.849	17.3	0.562	57.6	**10 x 10**	**W10**
54	80.36	1.49	12590	4304	11.1	6.49	983	338	1091	512	0.849	15.8	0.623	75.6		
60	89.29	1.50	14270	4858	11.2	6.55	1096	379	1224	575	0.850	14.5	0.717	102		
68	101.19	1.51	16390	5558	11.3	6.57	1241	432	1398	657	0.849	12.9	0.832	148		
72	107.15	1.52	17510	5928	11.3	6.59	1313	458	1484	697	0.848	12.4	0.898	172		
77	114.59	1.52	18950	6391	11.4	6.61	1408	494	1599	752	0.849	11.6	0.976	213		
88	130.96	1.54	22210	7436	11.5	6.67	1613	570	1850	869	0.850	10.3	1.16	314		
89	132.45	1.55	22530	7531	11.6	6.69	1631	576	1869	878	0.850	10.3	1.19	319		
100	148.82	1.56	25900	8598	11.7	6.73	1838	655	2127	999	0.851	9.3	1.38	452		
112	166.67	1.58	30000	9870	11.9	6.81	2075	744	2424	1137	0.851	8.5	1.63	626		
65	96.73	1.79	22250	7308	13.4	7.69	1445	479	1592	726	0.850	19.3	1.56	91.2	**12 x 12**	**W12**
72	107.15	1.80	24830	8117	13.5	7.72	1596	531	1767	806	0.850	17.6	1.75	122		
79	117.56	1.81	27670	9060	13.6	7.77	1760	589	1958	895	0.850	16.2	1.98	161		
87	129.47	1.82	30810	10030	13.7	7.80	1936	651	2164	990	0.851	14.9	2.22	212		
92	136.91	1.82	32820	10700	13.7	7.83	2048	692	2297	1053	0.851	14.2	2.39	249		
96	142.86	1.83	34670	11240	13.8	7.86	2148	728	2413	1106	0.853	13.6	2.53	285		
106	157.75	1.84	38750	12570	13.9	7.90	2369	808	2680	1230	0.851	12.5	2.87	378		
120	178.58	1.86	44560	14350	14.0	7.94	2675	917	3053	1399	0.851	11.2	3.34	537		
133	197.93	1.87	50900	16300	14.2	8.04	2995	1037	3440	1581	0.853	10.2	3.88	734		
136	202.39	1.87	51790	16560	14.2	8.02	3041	1052	3501	1606	0.852	10.1	3.95	771		
152	226.20	1.89	59610	18910	14.4	8.09	3424	1193	3975	1823	0.853	9.2	4.62	1074		
161	239.59	1.91	64200	20320	14.5	8.15	3643	1276	4247	1951	0.854	8.7	5.03	1271		
170	252.99	1.92	68490	21540	14.6	8.17	3844	1349	4502	2064	0.854	8.4	5.40	1480		
190	282.75	1.94	78870	24640	14.8	8.27	4318	1529	5105	2342	0.855	7.7	6.35	2034		
210	312.51	1.96	89270	27660	15.0	8.33	4779	1702	5703	2613	0.855	7.1	7.32	2693		
230	342.28	1.98	100600	30880	15.2	8.41	5261	1886	6332	2898	0.856	6.6	8.39	3490		
252	375.02	2.01	113200	34470	15.4	8.49	5783	2087	7020	3211	0.856	6.2	9.62	4495		
278	413.71	2.04	129600	39040	15.7	8.59	6437	2339	7890	3604	0.857	5.7	11.3	5953		
279	415.20	2.04	129500	39010	15.7	8.59	6435	2338	7888	3602	0.857	5.7	11.3	5946		
305	453.89	2.07	147600	43690	16.0	8.69	7122	2599	8807	4006	0.860	5.4	13.1	7719		
336	500.02	2.10	169000	49420	16.3	8.80	7913	2907	9882	4487	0.861	5.0	15.3	10110		

Note: Calculated based on 1kg = 2.2046 lb and 1 m = 3.2808 feet